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Email: [redacted]
Organization: BioMADE

General Comment

See attached file(s)

Attachments

BioMADE RFI on AI for Bioindustrial Processes - Final

BioMADE Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

The following response is from BioMADE and is being submitted by Andrew Eichenbaum, Sr. Director of Data Science at BioMADE, and Chris Dunlap, Director of Information Systems & Security at BioMADE.

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BioMADE is a non-profit Manufacturing Innovation Institute (MII) catalyzed by the U.S. Department of Defense (DoD). By supporting the development of biomanufacturing technologies, BioMADE and its network of over 300 members across 38 states are strengthening American competitiveness, creating a more resilient supply chain, re-shoring manufacturing jobs, and producing new bio-based products from American-grown feedstocks. BioMADE is also building a globally competitive Science, Technology, Engineering and Mathematics (STEM) workforce to ensure American workers are prepared and ready to fill new jobs within this rapidly growing industry.

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AI is critical to continued U.S. leadership in all manufacturing sectors. This includes bioindustrial manufacturing, which uses biological systems such as microbes (e.g., bacteria, yeast, and algae) to create new materials or secure alternatives to existing materials. BioMADE advances biomanufacturing technologies from early development in the laboratory through pilot scale-up to commercialization, with a focus on Manufacturing Readiness Levels (MRLs) 4-7. Data is a core component of this scale-up process, and AI will play a critical role in accelerating the commercialization of chemicals and materials made from biology. With our member companies and organizations, BioMADE is currently developing a Digital Backbone that will facilitate the collection, analysis, and sharing of bioindustrial data to rapidly grow the use of AI in this sector.

AI could have a transformative impact across all stages of bioindustrial manufacturing. Early in development, AI can facilitate the discovery and creation of microbes that are used for bioindustrial manufacturing, thereby enabling higher yields or lower costs of production. AI can also be used to automatically manage and optimize bioindustrial processes. For example, by analyzing water and energy usage, raw material consumption, and microorganism growth conditions, AI can help increase efficiency and reduce costs. In addition, AI-based optimization of supply chains, including through materiel demand forecasting, inventory management, and disruption predictions, will enable companies to more efficiently and robustly source the inputs they need to manufacture a range of bioproducts.

The major barrier to fully realizing these and other benefits of AI is a lack of data systems and standards that would enable the large-scale collection, analysis, and sharing of bioindustrial datasets. Making bioindustrial datasets truly AI ready will require four primary activities:

1. Development of a set of centrally created, managed, and maintained data systems such that bio-data, bio-analytics, and bio-process definitions, can be stored in a secure but easily shareable platform.
2. Creation of a set of standards for bio-data and bio-process management and tracking so that scientists and engineers across the U.S. can "speak the same language" and easily compare their data and processes.
3. Definition of a bio-data governance and security policy that can maximize ecosystem learning while protecting U.S. data from foreign adversaries.
4. A dedicated and focused education program around the collection, management, and analysis of data with specific courses focused on using AI in bioindustrial manufacturing.

These recommendations are supported by recent reports from the National Security Commission on Emerging Biotechnology[1] and National Academies[2].

1-2. Develop Data Systems and Standards

Biodata platform standards and best practices for data production and data storage will play a critical role in enabling the use of AI in bioindustrial manufacturing. Currently, there are a wide range of bioindustrial data formats and storage approaches. As a result, researchers can only leverage small, local datasets to inform their technology development. With the standardization of biodata formats for storage, scientists will be able to access, use, and understand the vast amounts of already generated data within the domestic biomanufacturing ecosystem. Coupled with a centralized, standardized, and secure data store, these standards will facilitate data sharing between industry, academic, and government partners and accelerate scale-up of products from the lab bench to commercialization. As the global leader in AI and biomanufacturing, the U.S. is well positioned to develop data standards and approaches that will be adopted worldwide.

3. Define Governance and Security Policies

All central data stores, standards, and sharing approaches must be created with an eye to governance and data security. Platforms should take a day-one approach to cyber security and incorporate traceability into their governance approaches to prevent and mitigate

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[1] National Security Commission on Emerging Biotechnology. 2023. *National Security Commission on Emerging Biotechnology Interim Report*. https://www.biotech.senate.gov/wp-content/uploads/2024/01/Biotech-Commission-Dec2023-Report.pdf

[2] National Academies of Sciences, Engineering, and Medicine. 2025. *Strategic Report on Research and Development in Biotechnology for Defense Innovation*. Washington, DC: The National Academies Press. https://doi.org/10.17226/27971.

attempts from adversaries to gain access to innovative technologies. AI-driven security tools for threat detection and anomaly analysis are essential to protect research, intellectual property, and critical infrastructure from cyber threats and biosecurity risks. U.S. leadership in AI – and in the software and data systems components necessary for the use of AI – will be critical to reduce U.S. dependence on foreign AI models that carry data privacy risks, risks of manipulation, and potential strategic vulnerabilities.

4. Create Education Programs

Education will be critical to enable the bioindustrial manufacturing workforce to incorporate AI into activities such as data collection, management, and analysis. Professional development programs will enable the current workforce to leverage AI in their existing manufacturing operations, including predictive maintenance, quality control, and workflow optimization. Programs should also be developed at the community college and university levels to train the next generation of technologists, scientists, and engineers. These programs should include real-world AI projects that apply theoretical concepts to practical scenarios, provide industry certifications, and job placement support. Instructors should be trained in the latest industry practices, including through externships that allow educators to gain hands-on experience in industrial settings and bring new skills back to the classroom. Sustained engagement with industry is important for collaborative curriculum development; provision or access to latest tools, technologies, or software used in the industry; and providing hands on learning opportunities for students and teachers.